



B.Tech. Degree IV Semester Examination April 2017

ME 15-1405 HYDRAULIC MACHINERY

(2015 Scheme)

Time: 3 Hours

Maximum Marks:60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Describe the Rayleigh's method for dimensional analysis.
- (b) What are the different types of similarities existing between model and prototype? Give a brief note on each one.
- (c) Why is draft tube used in a reaction turbine? Describe with neat sketch two different types of draft tubes.
- (d) What is cavitation? How can it be avoided in reaction turbines?
- (e) What are unit quantities? Define the unit quantities for a turbine.
- (f) Define specific speed of a centrifugal pump. Derive an expression for the same.
- (g) What is air vessel? Describe the function of air vessel for reciprocating pumps.
- (h) Draw and discuss operating characteristics of a centrifugal pump.
- (i) The pressure intensity of water supplied to an intensifier is 20 N/cm² while the pressure intensity of water leaving the intensifier is 100 N/cm². The external diameter of the sliding cylinder is 20 cm. Find the diameter of the fixed ram of the intensifier.
- (j) Obtain the expression for Rankine's efficiency of a hydraulic ram.

PART B

(4 × 10 = 40)

- II. (a) The frictional torque T of a disc of diameter D rotating at a speed N in a turbulent flow depends on fluid of viscosity μ and density ρ . Using Buckingham π theorem, obtain an expression for T . (6)
- (b) A pipe of diameter 1.5 m is required to transport an oil of specific gravity 0.90 and viscosity 3×10^{-2} Poise at the rate of 3000 litre/s. Tests were conducted on a 15 cm diameter pipe using water at 20°C. Find the velocity and rate of flow in the model. Viscosity of water at 20°C = 0.01 Poise. (4)

OR

- III. (a) A jet of water having a velocity of 30 m/s strikes a series of radial curved vanes mounted on a wheel which is rotating at 200 r.p.m. The jet makes an angle of 20° with the tangent to the wheel at inlet and leaves the wheel with a velocity of 5 m/s at an angle of 130° to the tangent to the wheel at outlet. Water is flowing from outward in a radial direction. The outer and inner radii of the wheel are 0.5 m and 0.25 m respectively. Determine (6)
 - (i) Vane angles at inlet and outlet.
 - (ii) Work done per unit weight of water.
 - (iii) Efficiency of the wheel.
- (b) Show that the efficiency of free jet striking normally on a series of flat plates mounted on the periphery of a wheel can never exceed 50%. (4)

(P.T.O.)

- IV. Design a Pelton wheel for a head of 80 m and speed 300 r.p.m. The Pelton wheel has to develop 103 kW of shaft power. (10)
Take $C_v = 0.98$, speed ratio = 0.45 and overall efficiency = 80%.

OR

- V. An inward flow reaction turbine has external and internal diameters of 1 m and 0.6 m respectively. The hydraulic efficiency of the turbine is 90% when the head on the turbine is 36 m. The velocity of flow at the outlet is 2.5 m/s and the discharge at the outlet is radial. If the vane angle at the outlet is 15° and width of the wheel is 100 mm at inlet and outlet, determine (10)
- (i) The guide blade angle.
 - (ii) Speed of the turbine.
 - (iii) Vane angle of runner at inlet.
 - (iv) Volume flow rate of turbine
 - (v) Power developed.

- VI. A centrifugal pump discharges $0.15 \text{ m}^3/\text{s}$ of water against a head of 12.5 m, the speed of the impeller being 600 r.p.m. The outer and inner diameters of the impeller are 500 mm and 250 mm respectively and the vanes are bent back at 35° to the tangent at the exit. If the area of the flow remains 0.07 m^2 from inlet to outlet, calculate, (10)
- (i) Manometric efficiency of the pump.
 - (ii) Vane angle at inlet
 - (iii) Loss of head at inlet to impeller when the discharge is reduced by 40% without changing the speed.

OR

- VII. Explain the effect of acceleration and friction in suction and delivery pipes on indicator diagrams. (10)
- VIII. Explain with the help of neat sketch, the principle and working of the following hydraulic devices: (10)
- (i) Hydraulic lift.
 - (ii) Hydraulic torque converter.

OR

- IX. (a) Explain the principle and working of a hydraulic crane with a neat sketch. (5)
(b) The efficiency of a hydraulic crane, which is supplied with water under a pressure of 70 N/cm^2 for lifting a weight through a height of 10 m, is 60%. If the diameter of the ram is 150 mm and velocity ratio is 6, find: (5)
- (i) The weight lifted by the crane.
 - (ii) The volume of water required in litres to lift the weight.